SaskPower's 200 MW Rose Valley Wind Facility Project

A Cost Benefit Analysis

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Purpose of The Cost-Benefit Analysis

The Government of Saskatchewan and SaskPower are advancing the Rose Valley Wind Energy Project, a proposed 200 MW utility-scale wind facility near Assiniboia, Saskatchewan, with construction planned for 2026–2027 and commercial operation expected in 2027 under a 30-year power purchase agreement. As Saskatchewan works to diversify its electricity system and reduce reliance on natural gas, policymakers must assess whether this project improves overall social welfare relative to continued fossil-based generation.

The purpose of this cost–benefit analysis (CBA) is to evaluate the full social costs and benefits of constructing and operating the Rose Valley Wind Project over its 30-year life. Following the Ten Steps of CBA, the project is compared to a baseline scenario in which electricity demand continues to be met by the existing generation mix. The analysis incorporates market impacts, non-market environmental effects, and distributional considerations to determine whether the project generates a positive net present value (NPV) at social discount rates of 3 percent, 5 percent, and 7 percent, and thus improves social welfare in Saskatchewan.

Wind Development vs. Status Quo

This analysis evaluates two alternatives to assess how the Rose Valley Wind Project affects social outcomes in Saskatchewan. The project option (Alternative A_1) involves constructing and operating a 200-MW wind facility, with capital costs incurred during construction and electricity generation beginning the following year over a 30-year project life. The baseline option (Alternative A_0) assumes the project is not built and that future electricity demand continues to be met primarily by natural gas generation. This baseline represents the most likely outcome in the absence of the project. Comparing these alternatives isolates the costs and benefits directly attributable to the wind project, including changes in operating costs, emissions, environmental damages, and local economic impacts.

Whose Benefits and Costs Count?

The primary referent group for this analysis is the residents of Saskatchewan, who bear most of the costs and receive most of the benefits of the Rose Valley Wind Project. Because SaskPower is a provincial Crown corporation, construction costs, operating expenses, and electricity pricing ultimately affect Saskatchewan households, businesses, and taxpayers and are therefore relevant to social welfare. Local communities near Assiniboia are also included due to direct impacts from land lease payments, municipal property taxes, and potential visual and noise effects. Nearby First Nations are included because the project may affect traditional land use, wildlife, and opportunities for economic participation. Although greenhouse gas reductions generate global benefits, they are included through the social cost of carbon. In contrast, financial gains accruing outside Saskatchewan are excluded unless they generate direct local spillover benefits. Municipal property tax revenues and land lease payments are classified as transfer payments within Saskatchewan, as they reallocate income between ratepayers, landowners, and municipalities but do not constitute net real resource costs or benefits to society as a whole.

Impact Categories and Measurement Indicators

Evaluating these alternatives requires identifying the full range of impacts the Rose Valley Wind Project may generate and determining how each can be measured. The primary costs of the project include the upfront capital expenditure for turbine installation and supporting infrastructure, as well as ongoing operation and maintenance (O&M) expenses over the 30-year project life. Additional costs include decommissioning at the end of the project's life and localized environmental impacts such as wildlife disturbance and visual or noise effects. Potential transmission or grid integration costs are also acknowledged, although these are not explicitly monetized due to data limitations.

The primary benefits of the project arise from the displacement of natural gas generation. These include avoided greenhouse gas emissions, improved local air quality through reductions in conventional air pollutants, and avoided variable operating costs from fossil fuel plants. The project also generates local economic flows, including land lease payments to landowners and municipal property tax revenues, which are transfer payments rather than real resource benefits, as they represent redistribution within Saskatchewan rather than changes in total social surplus.

Where market values exist, such as operating costs, land lease payments, and property tax revenues, monetary values are assigned directly using available data. For non-market effects, including avoided environmental damages and climate impacts, the analysis applies established valuation methods such as the social cost of carbon and per-unit damage estimates drawn from the literature and utility modelling guidance. Together, these impact categories and their associated measurement indicators form the basis for quantifying the project's overall contribution to social welfare.

Predicting Quantitative Impacts Over the Life of the Project

The physical impacts of the Rose Valley Wind Project are predicted over a 30-year operating life beginning in 2027. In the base case, annual electricity generation is estimated using an installed capacity of 200 MW, a capacity factor of 37 percent, and 8,760 hours per year, yielding approximately 651,000 MWh of electricity annually. This base-case capacity factor is derived from the five-year historical average of SaskPower's wind generation data, while bull and bear cases apply higher and lower capacity factor assumptions to reflect uncertainty in wind conditions and weather-related performance. Greenhouse gas impacts are calculated using a displaced emissions intensity of 0.338 tonnes of CO_2 per MWh, reflecting the displacement of combined-cycle natural gas generation and based on utility system modelling parameters provided by Professor Brett Dolter (2025). Annual operating quantities, including wind O&M activity, land lease obligations, property tax payments, wildlife impacts, and avoided fossil

variable O&M, are modeled on a year-by-year basis. A one-time noise disamenity effect is applied in the first year of operation, and a decommissioning impact is applied in the final year of the project life. These physical impact estimates form the basis for the monetary valuation undertaken in the following step.

Monetizing the Impacts

All quantified physical impacts are converted into monetary values using market prices, policy-based shadow prices, and established valuation methods from *Cost–Benefit Analysis: Concepts and Practice* by Boardman et al. (2018). The project's upfront capital cost is valued at $450 million in the construction year, based on cost estimates reported by SaskPower. Annual wind operation and maintenance costs are valued at $4.46 million per year, calculated using an O&M cost of $22,308 per MW-year from Dolter (2025) and scaled to the project's 200 MW capacity. Avoided greenhouse gas emissions are monetized using a carbon price of $125 per tonne of CO_2, corresponding to the federal minimum carbon pollution price in 2027 under Canada's carbon pricing benchmark, which increases by $15 per tonne annually from 2023 to 2030 (Government of Canada, 2023). This carbon price is applied to displaced emissions of 0.338 tonnes of CO_2 per MWh, based on utility system modelling parameters in Dolter (2025). Avoided non-carbon air pollution damages correspond to the "Avoided Damage" benefit stream in the model and are valued at $24 per MWh, using non-market damage estimates from Dolter (2025) and scaled by annual electricity generation. Avoided fossil variable operating costs are valued at $4 per MWh, reflecting short-run dispatch cost savings from displaced combined-cycle natural gas generation, again using modelling parameters from Dolter (2025).

Local fiscal and community impacts are monetized using Saskatchewan policy, municipal administrative frameworks, and project documentation. Annual property tax revenue of $651,525 is calculated using Saskatchewan's industrial assessment methodology and the Rural Municipality of Excel

No. 71 mill rate structure. Land lease payments are based on North American industry benchmark lease rates of $5,000, $6,500, and $8,000 per megawatt per year, as reported by LandApp (2025). Applied to the project's 200 megawatts of installed capacity, these values imply annual lease payments of $1.0 million in the bear case, $1.3 million in the base case, and $1.6 million in the bull case. In addition, an annual Community Benefits Fund contribution of $45,000 is included based on public commitments made by Potentia Renewables for the Rose Valley Wind Energy Project. Environmental disamenities are monetized using willingness-to-pay estimates from Boardman et al. (2018), with a one-time noise cost of $1.25 million applied in the first year of operation and an annual wildlife impact cost of $14,598 applied over the project life. A decommissioning cost of $22.5 million is applied in the final year of the project. Together, these monetary values form the localized stream of costs and benefits used in the present value analysis.

Discounting Benefits and Costs to Present Values

All future costs and benefits generated by the Rose Valley Wind Project are discounted to present values to reflect society's preference for receiving benefits sooner rather than later and for deferring costs into the future. Discounting is required because individuals place greater weight on near-term impacts than on distant future impacts, reflecting time preference and short-term decision making. In addition, funds invested in this project cannot be invested elsewhere, so discounting also reflects the opportunity cost of capital and the return that could be earned in alternative uses. Finally, uncertainty about future outcomes means that benefits and costs realized far in the future are inherently riskier than those realized today. This analysis applies three social discount rates of 3 percent, 5 percent, and 7 percent, consistent with standard practice in public-sector cost–benefit analysis. Using multiple discount rates allows the results to reflect uncertainty about the appropriate social rate of time preference and capital scarcity. Each annual net benefit is discounted back to the construction year using these rates to

generate a stream of present values that can be directly compared and aggregated in the NPV calculation.

Calculating NPV

The NPV of each alternative is calculated by summing the discounted net benefits over the 30-year life of the project. The baseline alternative (A_o), in which the Rose Valley Wind Project is not built, and electricity continues to be supplied using the existing natural gas-based system, is assigned an NPV of zero. This allows the project alternative (A_1) to be evaluated solely based on its incremental costs and benefits. For the project scenario, annual net benefits are calculated as total benefits minus total costs in each year and are then discounted back to the construction year using social discount rates of 3 percent, 5 percent, and 7 percent.

In the base case, the project yields NPVs of approximately $309.0 million at 3 percent, $147.1 million at 5 percent, and $33.1 million at 7 percent, indicating that benefits exceed costs under standard social discounting assumptions. In the bull case, NPVs increase substantially to $543.7 million at 3 percent, $327.5 million at 5 percent, and $175.6 million at 7 percent, reflecting more favorable assumptions for generation and revenues. In the bear case, the project remains viable at $178.6 million at 3 percent and $38.6 million at 5 percent but produces a negative NPV of −$60.8 million at the 7 percent discount rate, reflecting more conservative assumptions combined with a higher opportunity cost of capital. Because the baseline NPV is zero, the project's NPV directly measures the net gain in social welfare from constructing the Rose Valley Wind Project rather than maintaining the status quo.

As a supplementary indicator, the internal rate of return is 10.57 percent in the bull case, 7.73 percent in the base case, and 5.70 percent in the bear case.

Distributional Analysis (Party-by-Party Impacts)

Because property tax revenues and land lease payments are transfer payments rather than net additions to social surplus, they primarily affect the distribution of project benefits rather than the aggregate net present value. The distributional effects of the Rose Valley Wind Project are examined to identify how costs and benefits are shared across different groups within Saskatchewan. Local landowners receive direct financial benefits through annual land lease payments, while the surrounding municipality benefits from increased property tax revenue and community fund contributions. Saskatchewan ratepayers and taxpayers ultimately bear the project's capital and operating costs through SaskPower's rate base, but they also benefit from reduced long-run greenhouse gas emissions, avoided natural gas fuel costs, and improved local air quality. Nearby Indigenous communities may experience both potential economic participation opportunities and localized environmental or land use impacts. From a broader perspective, the climate benefits associated with reduced greenhouse gas emissions accrue beyond Saskatchewan as part of global emissions mitigation, while most financial costs and revenues remain within the province. Although the distribution of impacts varies across groups, the project generates a positive net social benefit in the base and bullish cases and also remains net beneficial in the bear case at lower social discount rates.

Sensitivity Analysis

A sensitivity analysis is conducted to test how robust the results of the cost–benefit analysis is to changes in key assumptions. Three scenarios are evaluated: a bull case, a base case, and a bear case. These scenarios vary the most uncertain and influential parameters in the model, including the wind capacity factor, avoided damage values per MWh, avoided fossil variable operating costs, land lease payments, and long-run project performance. The social discount rate is also varied across 3 percent, 5 percent, and 7 percent to reflect uncertainty in society's time preference and the opportunity cost of

capital. The results show that the project produces positive net present values across all discount rates in both the base and bull cases. In the bear case, the project remains net beneficial at the 3 percent and 5 percent discount rates but becomes slightly negative at the 7 percent discount rate. This indicates that the economic viability of the project is most sensitive to conservative assumptions combined with higher discounting, but overall, the results are robust under reasonable parameter values.

Final Recommendation

Based on the results of the cost–benefit analysis, the Rose Valley Wind Project is recommended for development under standard social discounting assumptions. In the base case, the project generates substantial positive net present value at 3 percent, 5 percent, and 7 percent discount rates, indicating that total social benefits exceed total social costs. The bull case shows even stronger net benefits, while the bear case remains positive at lower discount rates, suggesting that the project remains economically attractive under most plausible conditions. In addition to producing net social benefits, the project contributes to Saskatchewan's long-term emissions reduction goals, diversifies the provincial electricity system. Although some localized environmental and social costs exist, these impacts are relatively small compared to the large climate, fuel savings, and significant local fiscal redistribution generated over the project's life. Overall, the evidence indicates that proceeding with the Rose Valley Wind Project would enhance social welfare and represent an efficient allocation of scarce public resources in Saskatchewan.

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